

packages/graph-view/src/utils/layout.ts

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Overview

`packages/graph-view/src/utils/layout.ts` (lines 1-266) introduces a new layout utility. It defines default dimensions (`NODE_WIDTH` , `NODE_HEIGHT` , `LAYER_CLUSTER_WIDTH` , `LAYER_CLUSTER_HEIGHT` , `PORTAL_NODE_WIDTH` , `PORTAL_NODE_HEIGHT` – lines 15-20) and exports three layout helpers:

- `applyDagreLayout` (lines 30-79) – builds a Dagre graph, applies spacing heuristics, and returns positioned nodes.
- `applyForceLayout` (lines 94-189) – constructs a d3-force simulation, optionally clusters by community, and ticks synchronously to convergence.
- ELK helpers (lines 195-266) – `ELK_DEFAULT_LAYOUT_OPTIONS` (lines 195-204), `nodesToElkInput` (lines 206-228), and `mergeElkPositions` (lines 231-266).

Key changes

- **Imports*
 - added `dagre` and d3-force functions (`forceSimulation` , `forceLink` , `forceManyBody` , `forceCenter` , `forceCollide` , `forceX` , `forceY`) plus type imports for `SimulationNodeDatum` , `SimulationLinkDatum` , `Node` , `Edge` , and `ElkInput` .
- **Constants*
 - default node and cluster sizes exported.
- **`applyDagreLayout`**
 - synchronous layout for small graphs, using Dagre's `layout` method.
- **`applyForceLayout`**
 - synchronous force-directed layout for knowledge graphs, with optional community clustering via `forceX` / `forceY` .
- **ELK helpers*
 - convert `xyflow` nodes/edges to `ElkInput` and merge ELK-computed positions back onto the original nodes, preserving width/height for container nodes.

Impact

- Centralizes layout logic and constants, reducing duplication.

- Provides deterministic, reproducible layouts for small and knowledge-graph use cases.
- Exposes a clear API for future components to import layout functions.

Risks & follow-ups

- Verify that components previously using inline layout logic correctly import and use the new helpers.
- Add unit tests for edge cases such as empty node lists or missing `id` fields.
- Benchmark `applyForceLayout` on larger graphs to ensure the tick limit (≤ 300) is sufficient; consider exposing a configurable tick count if needed.
- Confirm that `nodesToElkInput` and `mergeElkPositions` handle container nodes as expected in downstream rendering.